



Society of Petroleum Engineers

SPE-228980-MS

Seeing Beyond the Curve: Coiled Tubing Hybrid Fiber Optics Technology and Engineered Retarded Acid Unlock Efficient Rig-Based Intervention in a Level-Four, Six-Legged Multilateral - A Landmark Execution from North Kuwait

Mohamed Hedi Slama, Kareem Badrawy, and Rustem Sunagatov, SLB, Kuwait City, Kuwait, KW.; Abrar Khaled Al-Osaimi, Kuwait Oil Company, Kuwait City, Kuwait, KW.; Tuleen Nazzal and Mohannad Adel Sebah, SLB, Kuwait City, Kuwait, KW.; Sergio Rondon Fajardo, SLB, Houston, TX, USA.; Ghassan Nicolas, SLB, Kuwait City, Kuwait, KW.; Mizan BinOmar Chong, Ghada Al-Mansour, Mohammad Al-Ajeel, Pravind Gupta Kumar, Priya Ranjan, AbdulSamad Ahmed, Abdulaziz Al-Subaie, Abdulaziz Najaf, Abdullah Al-Ajmi, Jassim Barki, and Saud Al-Haddad, Kuwait Oil Company, Kuwait City, Kuwait, KW.; Varun Mahajan, SLB, Kuwait City, Kuwait, KW.

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228980-MS](https://doi.org/10.2118/228980-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Advancements in horizontal and multilateral drilling have allowed the construction of extended-reach wells with increasingly complex architectures, enabling operators to maximize reservoir contact and production potential. However, these intricate well geometries—particularly level-four multilateral configurations—present significant challenges for rig-based intervention, particularly in post-drilling cleanout and stimulation operations. Coiled Tubing (CT) matrix acid stimulation remains a vital early intervention to restore well productivity and mitigate formation damage, yet traditional approaches often fall short in such highly complex environments.

In the North Kuwait Field, an operator implemented a cutting-edge stimulation strategy in a level-four, six-lateral multilateral oil producer targeting the Lower and Upper Tuba carbonate reservoirs. The well featured six 6 1/8-in. open-hole laterals, each averaging approximately 2,500 ft, and presented moderate reservoir conditions of 130°F bottomhole temperature and 2,100 psi pressure. The objective was to efficiently clean and stimulate multiple laterals using an integrated approach that minimized rig time, reduced operational complexity, and maximized fluid placement accuracy.

The selected intervention strategy utilized Coiled Tubing Hybrid Fiber Optics Technology (CT HFOT) combined with a multilateral entry tool (MLT) and a carefully engineered chemical treatment workflow. Real-time downhole measurements—including gamma ray, casing collar locator (CCL), CT internal pressure, annular pressure, annular temperature, and axial force—were streamed via CT HFOT to surface, enabling precise depth correlation, rapid lateral identification, and informed on-the-fly decision making during tool actuation and fluid placement.

Prior to acid stimulation, the three target laterals were cleaned and conditioned using a multifunctional solvent blend of aromatic hydrocarbons and surfactants, designed to disperse oil-based mud (OBM) residues, restore water-wet conditions, and prepare the formation for acid interaction without mechanical agitation. The subsequent matrix acid stimulation employed a Single-Phase Retarded Inorganic Acid System (SPRIAS), which demonstrated enhanced wormholing performance and superior placement efficiency compared to conventional emulsified acid systems typically used in carbonate formations. The integration of CT HFOT and SPRIAS chemistry enabled the profiling, cleanout, and stimulation of three laterals in a single CT run, with each lateral logged and accessed in under four hours. This represented a 60% reduction in total intervention time relative to traditional CT operations, driven primarily by real-time diagnostics, optimized tool actuation, and streamlined chemical treatment.

This case marks a landmark field execution in the Middle East and establishes a new standard for rig-based intervention in complex multilateral wells. The combination of hybrid fiber-optic telemetry, advanced multilateral entry techniques, and novel acid systems proved instrumental in enhancing operational efficiency, reducing non-productive rig time, and improving stimulation effectiveness in challenging carbonate environments. The success of this operation offers a compelling blueprint for similar interventions in high-complexity wells across the region.

Introduction — Sabriyah North Kuwait Tuba Formation - Case for Contact-Rich Development

The Upper Cretaceous Tuba Limestone in Sabriyah (Fig.1) sits between the Wara Formation below and the regionally extensive Ahmadi Shale above—an arrangement that provides both plumbing and seal for a tight but material carbonate target. Internally, Tuba is subdivided into Lower, Middle, and Upper packages (TL, TM, TU), with the Upper commonly parsed into TU10–TU90.

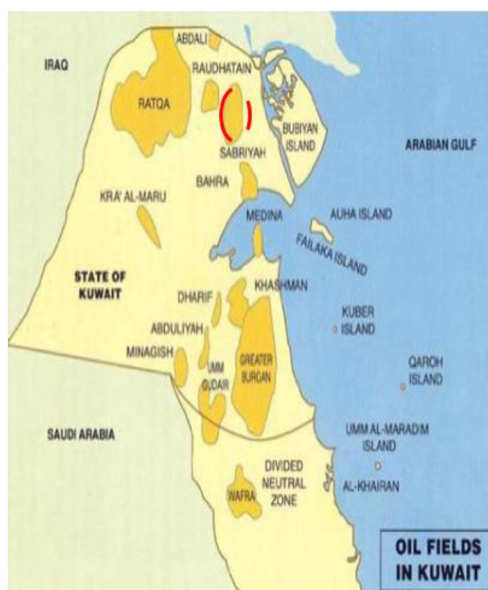


Figure 1—Sabriyah Field discovered in 1957

This stacking (Fig. 2) reflects deposition on a generally low-energy carbonate ramp punctuated by higher-energy inner- to mid-ramp pulses and quieter outer-ramp/basinal episodes that built shore-parallel geobodies later modified by strong diagenetic overprint. Thickness increases toward the northeast in Sabriyah, and the Upper Tuba locally behaves as a field-scale transmissibility barrier separating

hydrocarbon-bearing intervals—an architectural nuance that matters for well placement, completion design, and drainage strategy.

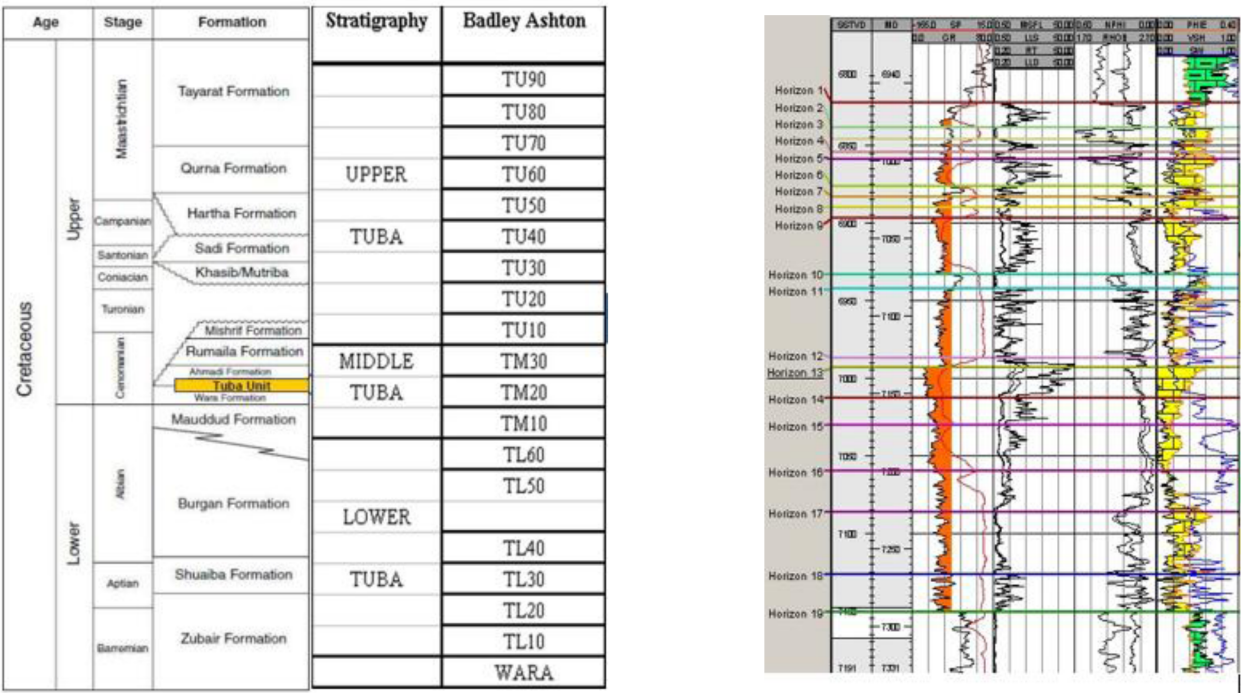


Figure 2—Tuba Sequence Stratigraphy and Reservoir Layering

Across Sabriyah, the core and lab evidence tell a consistent story. Detailed core description, thin-section petrography, Scanning Electron Microscopy (SEM) imaging, and Mercury Injection Capillary Pressure (MICP) collectively show that the original muddy and skeletal textures have not remained "as deposited." Early compaction and pressure-solution tightened parts of the rock; multiple pulses of calcite and dolomite cement locally plugged pores; later dissolution and natural fracturing reopened pathways; bioturbation and neomorphism further rearranged the fabric. The net result is a vertical and lateral patchwork of reservoir and non-reservoir layers where pore types and pore-throat sizes can flip over tens of feet. Permeability anisotropy and capillary behavior track those changes much more strongly than gross lithostratigraphy. In practical terms, flow units are thinner and more compartmentalized than log-only interpretations imply, and porosity-permeability transforms must be conditioned to facies and paragenesis rather than applied as a single field-wide relationship.

From a development vantage point, Tuba has long been the "sleeping" reservoir of North Kuwait: significant hydrocarbons in place, limited historical penetration, and early production that lagged what the rock could deliver when approached with generic workflows.

Recent integrated studies repositioned Tuba from a minor to an "upcoming" strategic target and highlighted that heterogeneity—driven by facies stacking and diagenesis—dominates performance. That heterogeneity must be honored in static and dynamic models, wellbore architecture, and surveillance plans. In practice, this means 3D characterization tied to core-calibrated petrophysics, realistic capillary/relative permeability curves by rock type, and an explicit treatment of internal baffles and barriers.

Two implications follow directly for well design. First, Tuba's bounding shales and internal barriers are a double-edged sword: they provide natural containment for stimulation and pressure management, but they also enforce flow compartmentalization. Lateral continuity within any single pay ribbon is inherently unreliable. Second, the controlling length scales for flow and pressure support are often shorter than a single

long horizontal can drain efficiently, particularly where diagenetic baffles and fracture corridors localize transmissibility (Fig. 3).

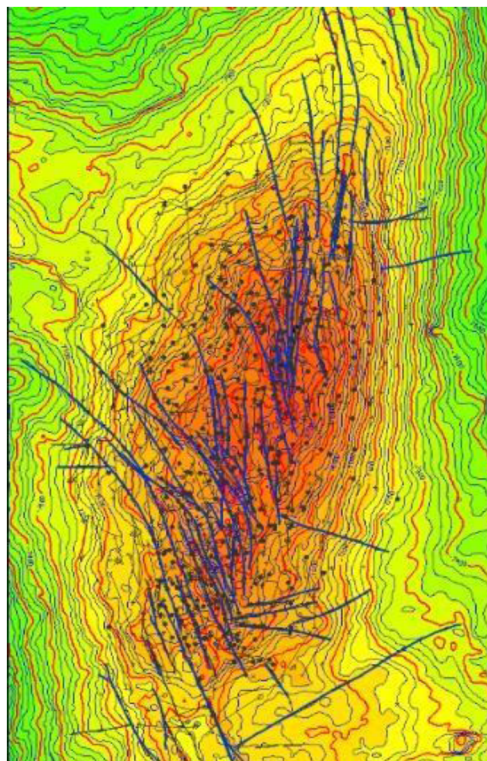


Figure 3—Some sealed faults have been interpreted at N/S.

In that context, development concepts that deliberately raise contact density and intersect multiple discrete flow units—rather than betting on vertical connectivity—are structurally advantaged. Put simply, Tuba's geology rewards geometries that multiply reservoir touchpoints and distribute drawdown across the TL/TM/TU stack, provided geosteering criteria and petrophysical cutoffs are tuned to the paragenetic model.

Finally, because North Kuwait development plans explicitly call out "upcoming reservoirs" as growth engines—with Tuba as a headline contributor—the technical aspect mandate is straightforward: design wells around the rock, not the other way around. For Tuba, that points toward Maximum Reservoir Contact (MRC) architectures in which multiple laterals are surgically placed into distinct flow ribbons to hedge against heterogeneity, elevate the probability of intersecting the best drains, and convert a geologically complex carbonate into a repeatable, manufacturing-style development.

Reservoir quality in Tuba is governed as much by diagenetic overprint as by original depositional fabric; successive phases of compaction, cementation, dissolution, neomorphism, pressure-solution, and natural fracturing have carved the formation into a fine-scale patchwork of pay and non-pay, with pore types and pore-throat networks changing over short distances. That heterogeneity—compounded by a tight matrix and variable fluid mobility—caps early-time productivity wherever drawdown is concentrated into a few dominant corridors. Smarter placement and more disciplined stage design help, but the decisive lever here is geometry. To translate discontinuous flow units into stable, commercial rates, contact density must rise and drawdown must be spread across more compartments. Multi-lateral MRC wells do exactly that—placing independent laterals into distinct TL/TU flow ribbons to raise encounter probability with the best petrofabrics, hedge against diagenetic baffles, and systematically expand swept volume.

Operating Multilaterals at Scale: Horizontal-Lateral Surveillance and Instrumented CT Access

Multilateral wells are now standard practice in Kuwait because they deliver the contact density the subsurface demands with fewer slots and a smaller surface footprint. The trade-off is operational complexity: once the rig leaves, surveillance and intervention become precision navigation under tight constraints. The task is no longer "run surveillance and clean out"; it is horizontal-lateral-aware access, confirmation, measurement, and control across a mainbore and several horizontal laterals that may differ in azimuth, inclination, completion status (barefoot vs. lined), and junction architecture (commonly TAML Level 2 or 4). Two decades of Middle East deployments have matured the hardware—modular junctions, through-completion window systems with landing profiles and sealbores, bending subs for horizontal-lateral entry, and isolation sleeves for lateral shut-off. These let you work without pulling the upper completion, but they demand precision: orientation, sealing, and pressure containment must be right the first time, or you miss the horizontal lateral or compromise junction integrity.

The physics change when switching from rotary drilling to CT. Rotary strings negotiate long drains because they rotate; CT does not. It moves linearly, accumulates friction quickly, and is more susceptible to helical buckling and lock-up especially after handing off through a junction into tighter radius or higher dogleg severity (DLS). A lateral that was "easy" while drilling can be marginal for CT unless you actively manage drag, fluid rheology, and junction differential limits. Access itself is non-trivial. CT multilateral access tools are typically flow-activated and indexed: map the window with a low-viscosity fluid, step through the indexing sequence to align the bending sub toward the targeted lateral, then kick the BHA into the horizontal lateral. Confirmation must be deterministic—look for a crisp CCL and GR signature paired with the characteristic pressure response that together prove you've crossed the junction and are advancing in the intended horizontal lateral. If indexing stalls or signatures are inconclusive, the professional response is to suspend operations, retrieve the BHA for function testing, and re-orient at surface; attempting to continue risks mis-entry into the wrong horizontal lateral, potential tool stranding, and days of avoidable non-productive time (NPT).

Surveillance planning also shifts from one-dimensional to horizontal-lateral-selective: depth correlation, hydraulics, and allowable rate/pressure windows must be set per lateral, and debris or fines near the window can localize bridges that jeopardize conveyance. Historically, trying to manage all of this from surface signals alone added runs and uncertainty. The step change is instrumented CT—embedding hybrid optical-fiber telemetry so bottomhole pressure/temperature, friction, and apparent weight are visible in real time. With live downhole data, you can tune rates below junction differentials, detect incipient buckling before lock-up, validate lateral connectivity with small rate or tracer pulses, and make informed go/no-go calls without chasing TD "just to be sure."

The latest fully electric multilateral tools (FEMLT) remove much of the ambiguity from horizontal-lateral re-entry. Instead of hydraulically "hunting" the window, an FEMLT uses electrically actuated mechanisms (bending sub and indexing mandrel) governed by surface-controlled software via the CT's embedded conductor/fiber. High-resolution angular encoders and tri-axial accelerometers/gyros provide real-time inclination and azimuth, while the toolstring integrates CCL and GR to confirm liner-shoe depth and lithology cues at the junction. In practice, the supervisor can dial the target inclination/azimuth in the control software; the tool orients to the commanded quadrant, verifies orientation against the downhole compass package, and then executes a controlled deflection into the selected horizontal lateral. Because the system runs closed-loop, it reports position, actuator status, and orientation quality in real time, and can auto-abort, re-index, or present a guided retry if tolerances are not met.

The net effect: mapping time drops from hours to minutes, indexing becomes repeatable rather than probabilistic, and mis-entry risk is sharply reduced. Additional advantages include (1) lower mapping-fluid exposure at the junction, (2) fewer surface cycles and pressure perturbations, (3) downhole self-diagnostics

that allow on-site fault isolation, and (4) cleaner digital records for post-job quality assurance and future re-entries.

Put together, this workflow is simple to state but exacting to execute: pre-job horizontal-lateral selection based on history and models; junction mapping and indexed entry with a bending sub; deterministic re-entry confirmation (CCL + GR + pressure); lateral-specific conveyance with real-time telemetry; optional isolation to segregate fluids when needed; then rapid cycling to the next horizontal lateral to compress the campaign. None of this makes multilaterals "easy," but it converts uncertainty into managed steps and short learning loops—the only way horizontal-lateral wells deliver their promised field-development efficiency at scale.

Single-Phase Retarded Inorganic Acid (SPRIAS) - deep dissolution without emulsion-related constraints

Carbonate stimulation is a controlled-dissolution problem: either remove near-wellbore damage by dissolving altered matrix or create conductive channels (wormholes) that connect the wellbore to effective permeability farther out. Hydrochloric acid (HCl) is the workhorse because it reacts rapidly with calcite and dolomite to form soluble Ca/Mg salts. Organic acids (formic, acetic) are useful when metallurgy or compatibility windows tighten, but their dissolving power is lower. Chelated systems can manage corrosion at higher temperatures, yet they trade away mineral dissolution capacity and add significant chemical cost. For years, emulsified acids—acid droplets dispersed in a hydrocarbon continuous phase—were the default way to slow reaction at elevated temperature. The emulsion reduces rock–acid contact frequency and pushes kinetics toward diffusion control, but it also introduces hydraulic and stability-management constraints: high fluid viscosity (driving up tubing/line friction), narrow mixing windows to keep the emulsion stable, sensitivity to shear/temperature history, and the ever-present risk of phase inversion or rag-layer formation.

SPRIAS was engineered to keep what matters and remove what doesn't. It preserves HCl's full carbonate solubility while using an inorganic retardation package to temper reaction kinetics. The fluid remains a single, aqueous phase with viscosity close to neat HCl, so hydraulics are cleaner: friction pressure is substantially lower than with emulsions, and the system stays compatible with standard friction-reducer chemistries. That buys rate flexibility within the same surface pressure envelope and avoids artificial slowdowns driven by friction—direct gains in placement quality and stage efficiency.

Mechanically, SPRIAS shifts dissolution toward a diffusion-limited regime at the rock face, spending less reactivity in the near-wellbore and carrying reactive strength deeper into the matrix. Core-flow programs at bottom-hole temperature repeatedly show pore volume to breakthrough (PVBT) (Fig. 4) for SPRIAS that is statistically similar to emulsified acid and markedly lower than plain HCl—evidence of deep penetration at practical injection rates without relying on an emulsion. The morphology is different, too: emulsions driven hard tend to initiate multiple competing wormholes and branch early, consuming extra acid to achieve outlet connection. SPRIAS more often develops a dominant, directed channel with fewer parasitic branches, delivering the same (or better) connectivity with less chemical spent.

Corrosion control and compatibility are handled without defaulting to chelants. Modern inhibitor packages, combined with the retarded kinetics, keep steel attack within limits across the intended thermal envelope while preserving HCl's dissolution power. With no hydrocarbon phase, field execution is simpler and safer: blending is straightforward, recipes are reproducible across pumps, and QA/QC is lighter—no emulsion-stability surveillance, no phase-inversion surprises, and far fewer shear/temperature sensitivities. The single-phase nature also trims inventory complexity and avoids emulsion-specific HSE burden around hydrocarbon handling.

Operationally, SPRIAS gives engineers the retardation of an emulsion with the hydraulics of neat acid. You achieve deeper, more selective channel development at the rates the placement model actually calls for, cut nonproductive mixing time, and keep the fluid envelope robust against real-world transients.

That combination explains its rapid uptake across Kuwait's carbonate assets: same dissolution objective, fewer emulsion-related constraints, and a cleaner, more repeatable operating window for high-temperature, heterogeneous reservoirs where hydraulic headroom and execution reliability are non-negotiable.

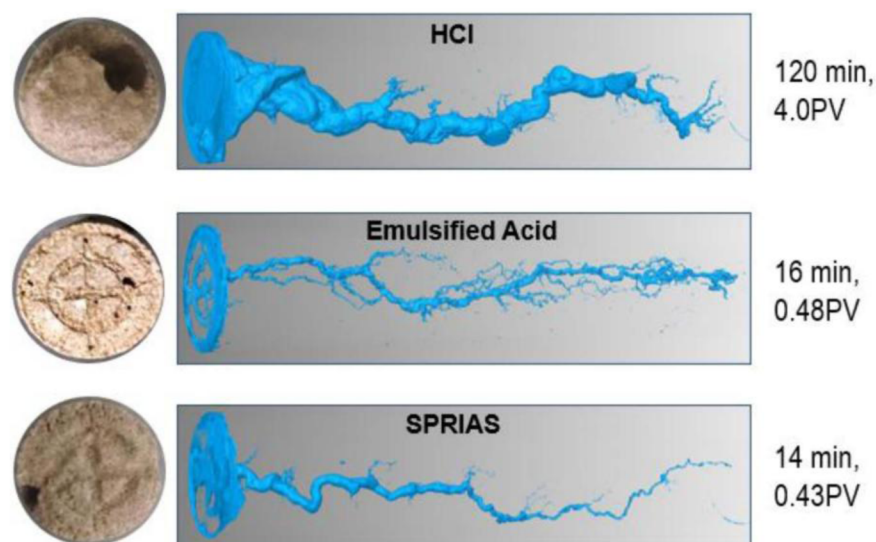


Figure 4—Representation of PVT experiment with 15% HCl, emulsified acid, and SPRIAS of similar HCl acid concentration on 1-in. diameter × 6-in. length, 8-md Indiana Limestone core at 250°F and 0.4 ml/min pumping rate – www.slb.com.

Case Study from Tuba Multilateral well in North Kuwait

In North Kuwait's Sabriyah field, the subject well (Fig.5) is a TAML Level-4, six-lateral multilateral intersecting the Tuba Upper (TU) and Tuba Lower (TL) carbonates. The target is a tight, compositionally complex reservoir with matrix permeability locally as low as ~10 mD, and interbedded shaly streaks were repeatedly observed while drilling—both factors that complicate placement and post-drill access.

This case study focuses on the upper three horizontals (L3–L5). The well was completed with 9 $\frac{5}{8}$ -in. production casing to approximately 8,400 ft MD, and a 7-in. production liner was cemented to the top of reservoir at ~9,000 ft MD. The mother bore (L3) is a 6 $\frac{1}{8}$ -in. barefoot lateral initiated at ~9,000 ft and drilled to ~11,500 ft MD, providing ~2,500 ft of exposure within TU. The L4 horizontal, also 6 $\frac{1}{8}$ -in., exited through a sidetrack window at ~9,600 ft MD and extended to ~12,300 ft MD, yielding ~2,700 ft of horizontal drain. The L5 horizontal (6 $\frac{1}{8}$ -in.) was kicked off higher from the L3 mother bore at the ~9,000-ft junction and drilled to ~11,800 ft MD, delivering ~2,800 ft of open-hole lateral. Bottom-hole conditions in TU were moderate, with BHT ~130 °F and BHP ~2,100 Psi.

The targeted Multi-lateral Level-4 horizontal well, currently underperforming in the Tuba upper and lower formations after 2 years of production was selected for matrix stimulation to bypass potential formation damage, lower reservoir performance and enhance productivity. A rig was called to pull out of the ESP string and prepare the well for coiled tubing intervention.

A 2in CT was used equipped with telemetry cable and FEMLT with direction and azimuth tool to navigate into the laterals directly. The intervention was designed to proof and drift the lateral bore three laterals to ensure accessibility then perform stimulation. The matrix simulation was designed to start with a preflush with aromatic solvent followed by stages of 200 Ft with spearhead acid of 15% HCl to initiate wormholing in the near wellbore then SPRIAS for delayed reaction resulting in deep penetration and Viscoelastic Diverting Agent (VDA) to fluid loss control then every 600 ft a stage of multimodal ball diverter is used to help diverting the acid in new area and enhancing the acid penetration.

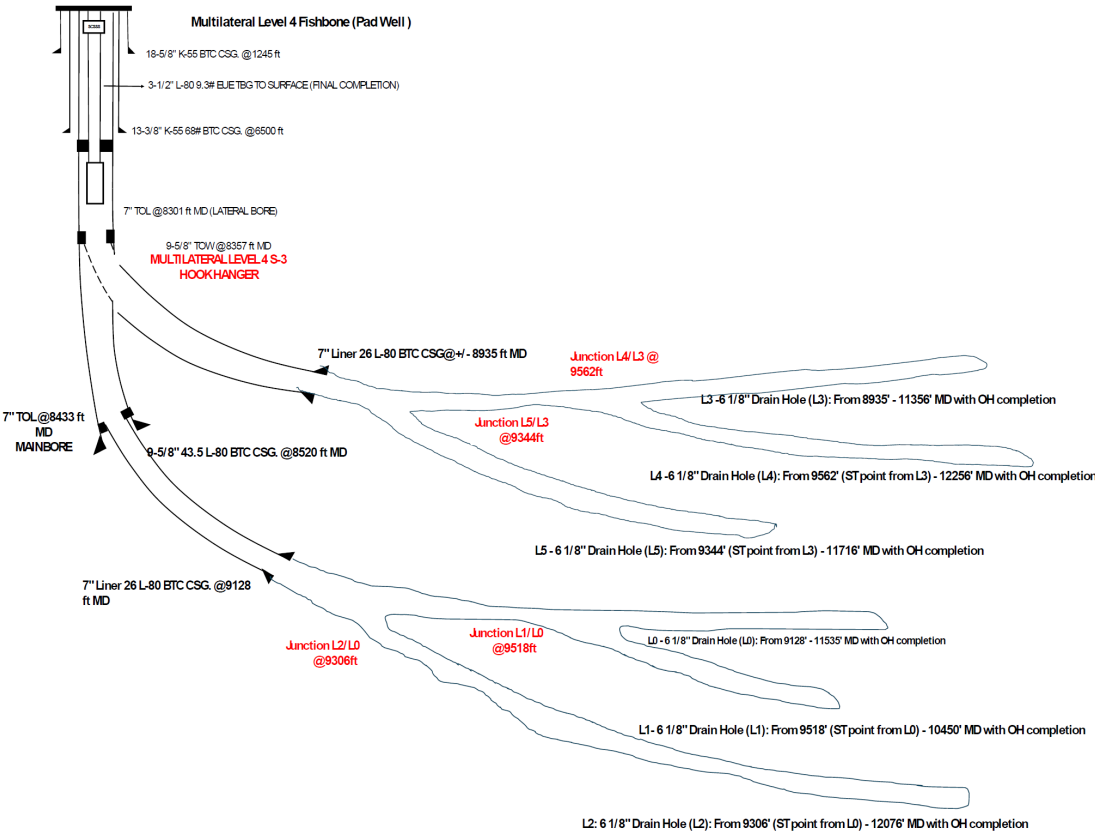


Figure 5—Wellbore Schematic

Table 1—Fluid pumping schedule.

Stage	Net Pay	CT Circulation Pressure	Max Squeeze Pressure	CT Movement Direction	Stage Fluid
	ft	(psi)	(psi)		
1	1,756	5,000	1,300	POOH	Pre-Flush
	1,756	5,000	1,300	RIH	Pre-Flush
2	200	5,000	1,300	POOH	15% Spearhead
		5,000	1,300	RIH	SPRIAS
		5,000	1,300	POOH	Viscoelastic Diverter
3	200	5,000	1,300	POOH	15% Spearhead
		5,000	1,300	RIH	SPRIAS
		5,000	1,300	POOH	Viscoelastic Diverter
4	200	5,000	1,300	POOH	15% Spearhead
		5,000	1,300	RIH	SPRIAS
		5,000	1,300	POOH	Viscoelastic Diverter
5		5,000	1,300	Stationary	Spacer
					Multimodal fibers diverter
					Spacer
6	200	5,000	1,300	POOH	15% Spearhead
		5,000	1,300	RIH	SPRIAS
		5,000	1,300	POOH	Viscoelastic Diverter

During intervention, CT didn't manage to reach total depth of the third lateral so decision was taken to start stimulation from the maximum depth reached. Stimulation was performed across 1,300 ft in L3 as shown in Fig. 6.

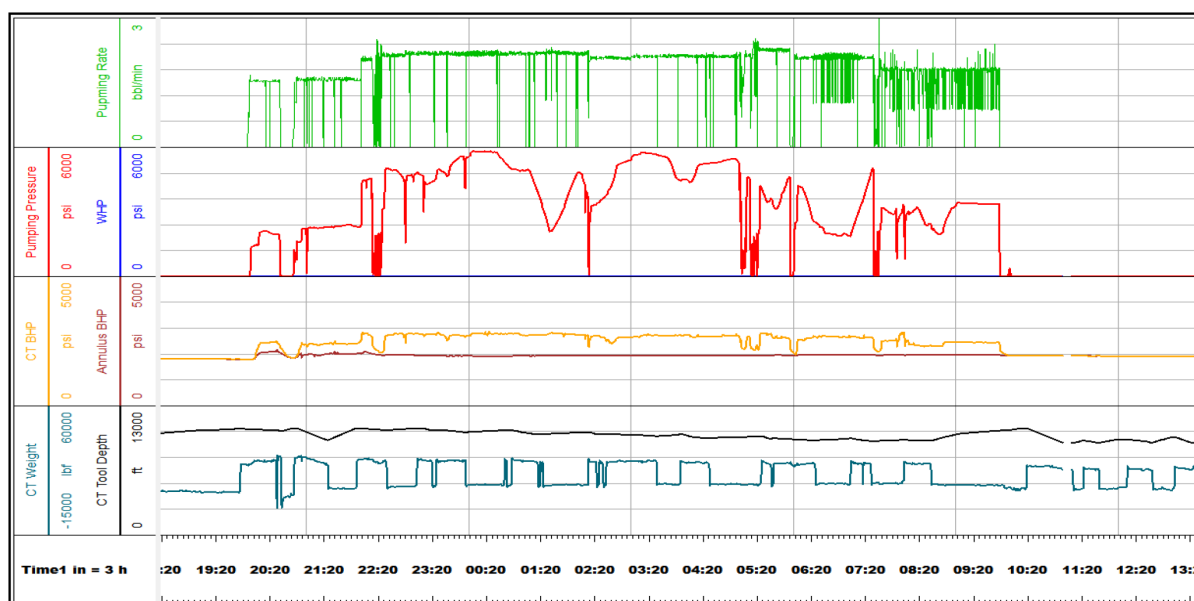


Figure 6—L3 Stimulation Acquisition Plot.

Once the treatment was done in L3, coiled tubing was pulled to the junction between L3 and L4 and using the FEMLT tool and the azimuth tool the coordinates for L4 was input for the tool and then access the next lateral. Once CT was at the TD of the lateral stimulation was performed across 2,700 Ft in L4 as shown in Fig. 7. The process was repeated again to access L5 as in Fig. 8.

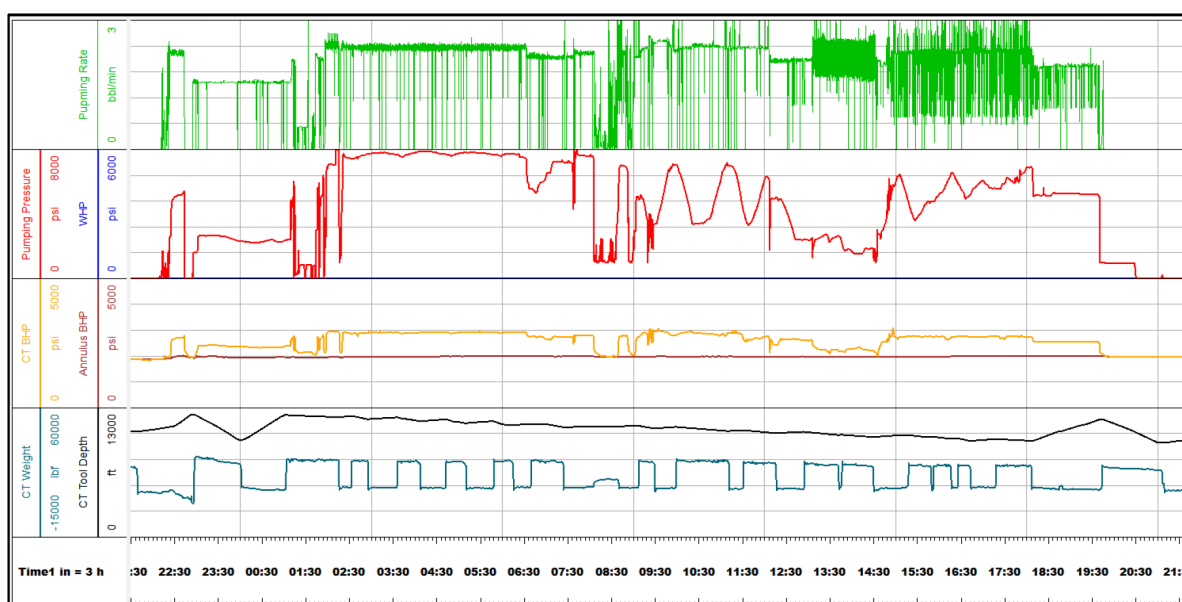


Figure 7—L4 Stimulation Acquisition Plot.

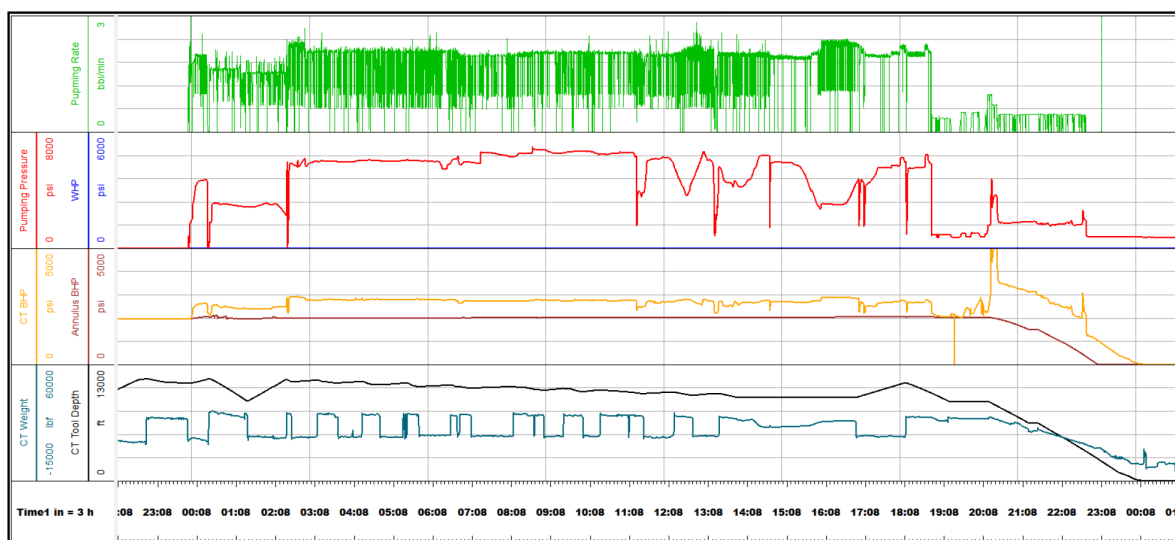


Figure 8—L5 Stimulation Acquisition Plot.

Conclusion and Way forward

Building on the case in the abstracta Level-4, six-lateral producer in Sabriyah's Tuba carbonates with ~2,500-Ft, 6⅛-in open-hole horizontals; BHT ~130 °F; BHP ~2,100 Psi—the differentiators were clear. We coupled Coiled Tubing Hybrid Fiber Optics Technology (CT HFOT) with a Fully Electrical Multilateral Tool (FEMLT) to make horizontal-lateral re-entry deterministic rather than probabilistic, and we replaced legacy emulsions with SPRIAS to retain deep dissolution while preserving HCl-class hydraulics. CT HFOT streamed GR, CCL, CT internal/annular pressures, annular temperature, and axial force in real time, enabling reliable depth correlation, rapid lateral identification, and on-the-fly actuation. A solvent pre-flush restored water-wet conditions and removed OBM residues; SPRIAS then delivered efficient matrix channels at practical rates. Net result: profiling, cleanout, and stimulation of three laterals in a single CT run, each lateral <4 h, for an intervention time ~60% lower than conventional CT. The points below distill the technical takeaways:

- **SPRIAS (chemistry):** The single-phase retarded acid achieved emulsion-class retardation with HCl-class hydraulics—low friction, simple blending, and deep, efficient wormholing at modeled rates. No emulsion stability management, no phase-inversion risk, and full compatibility with standard friction-reducer chemistries.
- **BHA (FEMLT + CT HFOT):** Surface-controlled FEMLT (indexing mandrel + bending sub) verified with CCL and GR, plus live BHP/BHT/pressure from CT HFOT, made lateral re-entry deterministic. Typical mapping/confirmation was 30–45 minutes per lateral with no TD passes required for confirmation.
- **Challenges and learnings:** Multilateral geometry and dogleg severity (DLS) constraints penalize guesswork. When signatures are inconclusive, the winning pattern is stop → function-test → re-index, not "press on." CT HFOT replaced surface inference with downhole facts, reducing buckling/lock-up risk and compressing cycle time and NPT.
- **Standardize the stack and KPIs:** Make FEMLT + CT HFOT the default access package and SPRIAS the baseline carbonate fluid. Track ≥90% first-pass re-entry, 30–45 minutes to map/confirm each lateral, and zero tool-driven NPT as non-negotiable metrics.
- **Govern by envelopes:** Screen SPRIAS locally (PVBT and corrosion). For each lateral, pre-set rate/pressure windows and DLS limits; require CCL + GR + pressure confirmation before placement. Maintain strict abort/re-index rules when downhole signatures drift.

- **Scale and evolve:** Campaign this workflow across Sabriyah and North Kuwait analogues with a simple dashboard (hours per lateral, re-entry %, SPRIAS rate adherence, coverage, NPT). Operator plan: embed the playbook in annual intervention programs. Provider roadmap: extend FEMLT temperature envelope, add auto-orientation and health diagnostics, and integrate richer downhole sensing to further shorten mapping and improve placement fidelity.

To convert this success into standard practice, we will formalize three enablers—FEMLT (Fully Electrical Multilateral Tool), CT HFOT (Coiled Tubing Hybrid Fiber Optics Technology), and SPRIAS—inside a horizontal-lateral-selective, gate-driven, metrics-anchored execution framework. FEMLT + CT HFOT becomes the default multilateral access package: FEMLT with a bending sub and integrated CCL/GR for orientation and depth control, with entry decisions governed by live BHP/BHT/pressure and orientation-quality factors streamed through CT HFOT. Performance targets are explicit and non-negotiable: $\geq 90\%$ first-pass re-entry, 30–45 minutes to map/confirm each horizontal lateral, and zero tool-related NPT. SPRIAS is adopted as the baseline carbonate chemistry for TU/TL, with emulsions retired unless screening fails; local validation is by lab tests and corrosion testing at representative conditions, and operating rates are set to exploit SPRIAS's low friction to meet modeled placement. The operating playbook is codified: pre-job lateral ranking from pressure/temperature history and modeled drawdown, with DLS and junction pressure limits set per lateral; in-job deterministic confirmation (CCL + GR + pressure), CT-based buckling watch, and hard abort/re-index criteria when signatures are inconclusive; post-job digital packs (orientation logs, CCL/GR overlays, BHP/BHT/pressure traces) fed back to the reservoir model.

We then scale across Sabriyah and analogue North Kuwait carbonates, governed by a simple dashboard (hours per lateral, re-entry success, SPRIAS rate adherence, coverage, NPT) to enforce learning cycles. The result is predictable execution—FEMLT + CT HFOT for precise, fast access and SPRIAS for deep, rate-stable dissolution—making "three laterals in one CT run, <4 h per lateral" a deliverable, not a headline.

Acknowledgments

The authors gratefully acknowledge the Kuwait Oil Company (KOC) Leadership Team for their strategic sponsorship and guidance. Their support enabled the planning, execution, and dissemination of this work.

We thank the Well Surveillance Rig Unit and the Field Development Team—North Kuwait, Sabriyah for their close collaboration in the field. Their operational expertise, data stewardship, and readiness to pilot new workflows materially improved the quality and relevance of the study.

We also acknowledge the SLB Reservoir Performance Intervention Technical Team for their technical contributions and disciplined support across planning, diagnostics, and intervention operations. Their input strengthened the analysis and helped translate lessons learned into actionable practice.

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